_A 5-bit 40x40 image has the following intensities, the equalized histogram for the gray level 15 is

Gray level	11	15	10	20	18
Count	580	320	150	250	300

a. 20 ✓

b. 15

c. 3

The correct answer is: 20

_After digitization process a digital image with M rows and N columns and L, intensity levels. the number of bits required to store this image is

a. M*N*log2(L) ✓

b. M*M*L

c. M*N*L

The correct answer is: M*N*log2(L)

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The image shown indicates a visual problem; which subprogram helps enhance it



a. B)Mat src = imread("x.jpg", 0);
```
Mat dst;
src.convertTo(src, CV_32F);
src = src + 1;
log(src, dst);
normalize(dst, dst, 0, 255,NORM_MINMAX);
convertScaleAbs(dst, dst);
```

b. C) Mat src=imread("x.jpg", 0);
```
Mat dst = src;
for (int i = 0; i < src.rows; i++)
    for (int j = 0; j < src.cols; j++)
        dst.at<uchar>(i, j) = 255 - src.at<uchar>(i, j);
```

c. A) Mat src = imread("x.jpg", 0); ✓
```
Mat dst;
equalizeHist(src, dst);
```

Your answer is correct.

The correct answer is: A) Mat src = imread("x.jpg", 0);
```
Mat dst;
equalizeHist(src, dst);
```

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Question 7

Correct

Mark 1.00 out of 1.00

Flag question

_For a continuous image f(x, y), Quantization is defined as

a. Digitizing the coordinate values
b. None of the mentioned.
c. Digitizing the amplitude values ✓

The correct answer is: Digitizing the amplitude values

Question 8

Correct

Mark 1.00 out of 1.00

Flag question

_For a continuous image f(x, y), sampling is defined as

a. Digitizing the amplitude values
b. None of the mentioned.
c. Digitizing the coordinate values ✓

The correct answer is: Digitizing the coordinate values

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Question **4**

Correct

Mark 1.00 out of 1.00

Flag question

_An image is defined as a function of f(x,y), where x,y represents .....and f represent.....

a. spatial coordinates , amplitude ✓

b. brightness , position

c. intensity , resolution

The correct answer is: spatial coordinates , amplitude

Question **5**

Complete

Not graded

Flag question

_Bit plane slicing is a method of representing an image with different planes which plane represent the most significant data. Show its content for pixel (3,7) in the following image matrix.

a. Most significant, 1

b. Least significant, 0

c. non of the mentioned

The correct answer is: non of the mentioned

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Question 9

Correct

Mark 1.00 out of 1.00

Flag question

_For the following image matrix, which of the following represent the enhancement effect at pixel (5,1) which used to enhance white or grey detail embedded in dark regions of an image.

```
[255, 218, 282, 285, 284, 255, 255, 255, 253, 255,
 253,  42,   2,   1,   0, 252, 254, 253, 255,  96,
 254,  49,   1,   0,   3, 255, 254, 253,  83,   0,
 255,  51,   0,   0,   0, 255, 253,  76,   0,   1,
 254,  47,   0,   6,   0, 255,  45,   0,   0,   0,
 255,  54,   0,   0,   2, 249, 143, 155, 155, 155,
 255, 255, 255, 255, 255, 255, 143, 155, 155, 155]
```

a. 3

b. non of the mentioned ✓

c. 51

The correct answer is: non of the mentioned

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Question 10

Complete

Not graded

Flag question

_For the following image matrix, which of the following represents the enhancement effect at pixel (2,3) used to enhance an image with low brightness?

```
[255, 218, 282, 285, 284, 255, 255, 255, 253, 255,
 253,  42,   2,   1,   0, 252, 254, 253, 255,  96,
 254,  49,   1,   0,   3, 255, 254, 253,  83,   0,
 255,  51,   0,   0,   0, 255, 253,  76,   0,   1,
 254,  47,   0,   6,   0, 255,  45,   0,   0,   0,
 255,  54,   0,   0,   2, 249, 143, 155, 155, 155,
 255, 255, 255, 255, 255, 255, 143, 155, 155, 155]
```

a. 32

b. 51

c. 0

The correct answer is: 0

Question 11

Correct

Mark 1.00 out of 1.00

Flag question

_the first and foremost step in Digital Image Processing is

a. Image rotation

b. Image acquisition ✓

c. Affine transformation

The correct answer is: Image acquisition

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Question 12

Correct

Mark 1.00 out of 1.00

Flag question

_The following image matrix represents the pixel values of an 8-bit image. Which of the following represent N4 and ND neighbors for pixel (3,4), respectively?

```
[255, 218, 282, 285, 284, 255, 255, 255, 253, 255,
 253,  42,   2,   1,   0, 252, 254, 253, 255,  96,
 254,  49,   1,   0,   3, 255, 254, 253,  83,   0,
 255,  51,   0,   0,   0, 255, 253,  76,   0,   1,
 254,  47,   0,   6,   0, 255,  45,   0,   0,   0,
 255,  54,   0,   0,   2, 249, 143, 155, 155, 155,
 255, 255, 255, 255, 255, 255, 143, 155, 155, 155]
```

a. {255,249,0,45}-{0,253,3,143}

b. {0,2,6,255}-{0,255,0,249}

c. {3,0,0,255}-{0,255,6,255} ✓

The correct answer is: {3,0,0,255}-{0,255,6,255}

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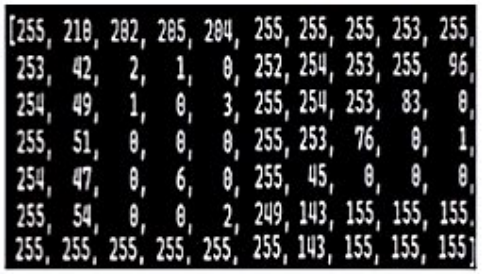
Question 13

Correct

Mark 1.00 out of 1.00

Flag question

_The following image matrix represents the pixel values of an 8-bit image. Which of the following represents the D_m distance between pixels p (1,2) and q (4,6), given the connectivity set ξ={0,1,2,3,6}



```
[255, 218, 282, 285, 284, 255, 255, 255, 253, 255,
 253,  42,   2,   1,   0, 252, 254, 253, 255,  96,
 254,  49,   1,   0,   3, 255, 254, 253,  83,   0,
 255,  51,   0,   0,   0, 255, 253,  76,   0,   1,
 254,  47,   0,   6,   0, 255,  45,   0,   0,   0,
 255,  54,   0,   0,   2, 249, 143, 155, 155, 155,
 255, 255, 255, 255, 255, 255, 143, 155, 155, 155]
```

a. 4

b. non of the mentioned ✓

c. 2.5

d. 3

The correct answer is: non of the mentioned

Question 14

Correct

Mark 1.00 out of 1.00

Flag question

_The method which increases the low dynamic range of the gray levels in a low contrast image to cover the full range of gray levels in order to improve image contrast is.

a. Histogram equalization ✓

b. Image smoothing

c. Contrast stretching

The correct answer is: Histogram equalization

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Question 15

Correct

Mark 1.00 out of 1.00

Flag question

_Which of the following represent transformation function of histogram equalization

a. S = c * log (1 + r)

b. S=L-r

c. S=(L-1)*CDF(r) ✓

The correct answer is: S=(L-1)*CDF(r)

Question 16

Correct

Mark 1.00 out of 1.00

Flag question

_Which of the following used to enhance the problem of shifting an object's location?

a. Image smoothing

b. Image translation ✓

c. Image skewing

The correct answer is: Image translation

Finish review

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_Digital image processing:

a. It describes the ratio between the maximum and minimum intensities in the Image
b. It is Processing of image data for storage, transmission and representation for autonomous machine perception ✓
c. It is a graph, which gives you an overall idea about the intensity distribution of an image.

The correct answer is: It is Processing of image data for storage, transmission and representation for autonomous machine perception

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